

Name: _____ Partner: _____ Period: _____

Context: Solving Trigonometric Equations

Trigonometric equations arise whenever a periodic model is set equal to a specific output value. In this activity your group will solve trig equations and inequalities in several settings. Choose **one tier** below based on your readiness. Work together — everyone should be ready to explain your group's reasoning.

Tier R — Remediate (Key)

The general solution of $\sin \theta = \frac{1}{2}$ is already provided:

$$\theta = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad \theta = \frac{5\pi}{6} + 2\pi k, \quad k \in \mathbb{Z}.$$

1. Family 1 solutions:

$$k = 0: \theta = \frac{\pi}{6} \quad k = 1: \theta = \frac{\pi}{6} + 2\pi = \frac{13\pi}{6} \quad k = 2: \theta = \frac{\pi}{6} + 4\pi = \frac{25\pi}{6}$$

2. Family 2 solutions:

$$k = 0: \theta = \frac{5\pi}{6} \quad k = 1: \theta = \frac{5\pi}{6} + 2\pi = \frac{17\pi}{6} \quad k = 2: \theta = \frac{5\pi}{6} + 4\pi = \frac{29\pi}{6}$$

3. Solutions in $[0, 4\pi]$: $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$ (4 solutions, since $4\pi \approx 12.57$ and $\frac{25\pi}{6} \approx 13.09 > 4\pi$).
How many? **4**

4. $\theta = \frac{\pi}{2} + \pi k$ in $[0, 2\pi]$: $\theta = \frac{\pi}{2}$ **and** $\theta = \frac{3\pi}{2}$

5. **Trig functions are periodic, so the same output value repeats once per period. In $[0, 4\pi]$ there are two full periods, giving two occurrences of each solution family.**

Tier A — Apply (Key)

1. $2 \sin \theta + \sqrt{3} = 0$:

Step 1: $\sin \theta = -\frac{\sqrt{3}}{2}$

Step 2: **Principal:** $\arcsin(-\frac{\sqrt{3}}{2}) = -\frac{\pi}{3}$ (Q4 reference = $\frac{\pi}{3}$). In $[0, 2\pi)$: $\theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$ (Q3) and $\theta = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$ (Q4).

Step 3: **Sine is negative in Q3 and Q4.**

Step 4: $\theta = \frac{4\pi}{3}$ and $\theta = \frac{5\pi}{3}$

2. $2 \cos^2 \theta - \cos \theta - 1 = 0$:

Factor: $(2 \cos \theta + 1)(\cos \theta - 1) = 0$

$u = \cos \theta$: $u = -\frac{1}{2}$ or $u = 1$

For $\cos \theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$

For $\cos \theta = 1$: $\theta = 0$

All solutions in $[0, 2\pi)$: $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

3. $d(t) = 2 \cos(\frac{\pi}{6}t) + 4 = 3$:

(a) $\cos(\frac{\pi}{6}t) = -\frac{1}{2}$. Let $u = \frac{\pi}{6}t$: $u = \frac{2\pi}{3}$ or $u = \frac{4\pi}{3}$. Then $t = \frac{6}{\pi} \cdot \frac{2\pi}{3} = 4$ and $t = \frac{6}{\pi} \cdot \frac{4\pi}{3} = 8$. **Solutions:** $t = 4$ and $t = 8$ hours.

(b) **The general solution gives $t = 4 + 12k$ and $t = 8 + 12k$. For $k = 1$: $t = 16$ and $t = 20$, both outside $[0, 12]$. Context restricts t to one 12-hour period, so only $t = 4$ and $t = 8$ are valid.**

Tier E — Extend (Key)

1. $f(\theta) = 0$ at $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

2. $f(\theta) > 0$ on $[0, \frac{\pi}{3}) \cup (\frac{5\pi}{3}, 2\pi]$ in $[0, 2\pi]$.

Teacher Note

From the table: $f(0) = 1 > 0$, $f(\pi/3) = 0$, $f(\pi/2) = -1 < 0$, $f(\pi) = -3 < 0$, $f(3\pi/2) = -1 < 0$, $f(5\pi/3) = 0$, $f(2\pi) = 1 > 0$. Positive only near 0 and 2π .

3. $2 \cos \theta - 1 > 0 \Rightarrow \cos \theta > \frac{1}{2}$.

Cosine $> \frac{1}{2}$ in $(-\frac{\pi}{3}, \frac{\pi}{3})$ per period. In $[0, 2\pi]$: $\cos \theta > \frac{1}{2}$ on $[0, \frac{\pi}{3}) \cup (\frac{5\pi}{3}, 2\pi]$. This confirms the table-based answer.

4a. $L(t) \geq 13.5$:

$3 \sin(\frac{2\pi}{365}(t - 80)) + 12 \geq 13.5 \Rightarrow \sin(\frac{2\pi}{365}(t - 80)) \geq \frac{1}{2}$.

Let $u = \frac{2\pi}{365}(t - 80)$. **Boundary:** $u = \frac{\pi}{6}$ and $u = \frac{5\pi}{6}$.

$t = 80 + \frac{365}{2\pi} \cdot u$.

$t_1 = 80 + \frac{365}{2\pi} \cdot \frac{\pi}{6} = 80 + \frac{365}{12} \approx 80 + 30 = 110$ (day 110).

$t_2 = 80 + \frac{365}{2\pi} \cdot \frac{5\pi}{6} = 80 + \frac{365 \cdot 5}{12} \approx 80 + 152 = 232$ (day 232).

$L(t) \geq 13.5$ for approximately days $110 \leq t \leq 232$.

4b. **This represents roughly late April through mid-August, when the city has more than 13.5 hours of daylight. The general solution produces values like $t = 110 + 365k$ for other years ($k \neq 0$), but since $t \in [1, 365]$ represents a single year, only $k = 0$ is contextually valid.**